

- (c) Plastic carrier bags are made from polythene. Each plastic carrier bag can take 500-1000 years to decompose and may never break down in landfill. Paper bags are not necessarily an environmentally friendly alternative. Manufacturing paper bags wastes a lot of natural resources. Even starch-based biodegradable bags use natural resources during their manufacture.

Supermarkets give customers a choice of buying single-use or re-usable polythene carrier bags.

Table 2 shows the number of both types of plastic bag sold in UK supermarkets from 2011 to 2013.

Year	2011	2012	2013
	Number of bags (millions)		
Single-use bags	7977	8079	8455
Re-usable bags	415	408	445

Table 2

- (i) State **one** environmental problem related to the disposal of **all** types of carrier bag. [1]

.....

- (ii) Calculate the percentage of plastic bags sold in 2013 that were single-use bags. [2]

Percentage = %



Examiner
only

(iii) Although more plastic carrier bags were sold in 2013 than 2012, the total **mass** of those bags changed from 70 400 tonnes to 67 300 tonnes.

Put a tick (✓) in **two** boxes next to statements which could explain the reason for the change in mass. [2]

the bags were made the same thickness but from a less dense plastic

☐

customers re-used their plastic bags more often

☐

the bags were made from the same plastic but were thicker

☐

the bags were made from the same plastic but were thinner

☐

the bags were made the same thickness but from a more dense plastic

☐

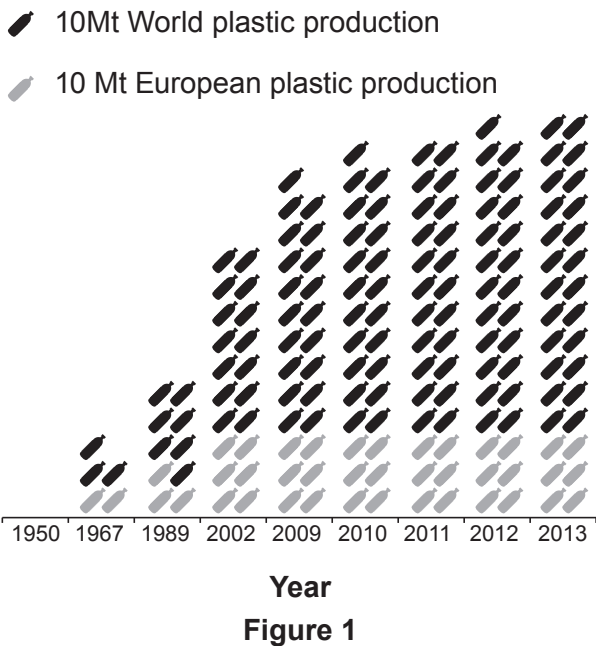
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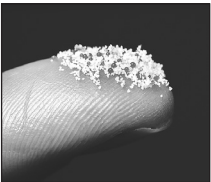
(b) **Microplastics in the Ocean**

Plastics are used in many areas of modern life. **Figure 1** shows the amount of plastic production between 1950 and 2013.

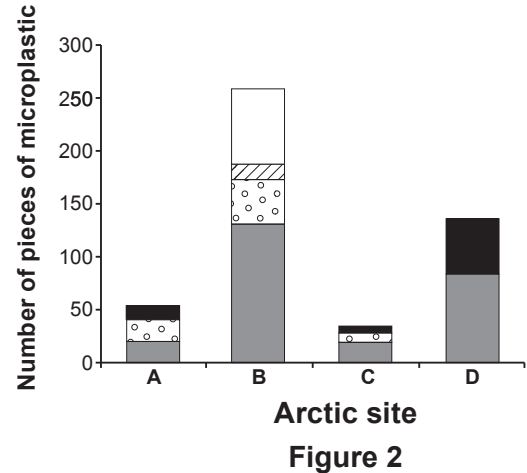


One of plastic’s greatest properties, its durability, is also one of the main reasons that plastics present a threat to the marine environment.

The term ‘microplastic’ is used to describe plastic particles that are less than 5 millimetres in diameter, which includes particles as small as 10 nanometres. Microplastics can be found in some cosmetic products, toothpastes and soaps.



Microplastics are spread throughout the oceans and are found on shorelines from the Arctic to Antarctica. **Figure 2** shows the number of microplastic pieces found in sea ice at four Arctic sites during a survey in 2014.



- Key Plastic**
- nylon
 - polyester
 - rayon
 - polypropylene
 - others

Microplastics have been found inside the bodies of marine animals. Microplastics often contain chemicals that can absorb poisons such as pesticides from the surrounding seawater. There is strong evidence of transfer of poisons from eaten microplastics into animal tissues. Nano-size microplastics have been shown to cross cell membranes, under laboratory conditions, causing tissue damage.

Public awareness of the potential for microplastics to damage marine animals is low compared with that of the impact of plastic litter in our seas and oceans. Effective education of society is essential to raise awareness of the damaging effects of microplastics.

